

OPTIMIZING TRANSFORMERS: INITIALIZATION AND HYPERPARAMETERS FOR MODULAR ARITHMETIC AND PERMUTATIONS

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ABSTRACT

We explore the impact of various initialization schemes and hyperparameters on Transformer models for modular arithmetic and permutation tasks, which are essential for understanding neural networks’ generalization on small algorithmic datasets. The challenge lies in identifying initialization methods and hyperparameters that facilitate faster and more effective grokking, where models initially overfit but eventually generalize well. We propose a comprehensive evaluation framework that includes different initialization schemes such as Xavier, He, Uniform, and Gaussian, along with varying learning rates, batch sizes, and the number of layers. Our contributions include a detailed analysis of how these factors influence training stability, speed to grokking, and final validation accuracy. We validate our approach through extensive experiments, showing that certain initialization schemes and hyperparameters significantly enhance the model’s ability to generalize.

1 INTRODUCTION

Understanding the generalization capabilities of neural networks on small algorithmic datasets is a critical area of research. In this paper, we focus on the performance of Transformer models when applied to modular arithmetic and permutation tasks. These tasks are fundamental in theoretical computer science and serve as benchmarks for evaluating the effectiveness of neural network architectures in learning and generalizing from limited data.

The challenge in these tasks lies in the phenomenon known as “grokking,” where models initially overfit the training data but eventually learn to generalize well (Power et al., 2022). Identifying the right initialization schemes and hyperparameters that facilitate faster and more effective grokking is non-trivial. This difficulty is compounded by the need to balance training stability, speed to grokking, and final validation accuracy.

To address these challenges, we propose a comprehensive evaluation framework that includes various initialization schemes such as Xavier (Glorot & Bengio, 2010), He (He et al., 2015), Uniform, and Gaussian, as well as different learning rates, batch sizes, and the number of layers. Our contributions are as follows:

- We provide a detailed analysis of how different initialization schemes and hyperparameters affect the training stability, speed to grokking, and final validation accuracy of Transformer models.
- We introduce a robust experimental setup that allows for the systematic evaluation of these factors across multiple tasks.
- We validate our approach through extensive experiments, demonstrating that certain initialization schemes and hyperparameters significantly enhance the model’s ability to generalize.

We verify our solution through a series of experiments that measure the training and validation performance of Transformer models under different initialization schemes and hyperparameters. Metrics such as training stability, speed to grokking, and final validation accuracy are used to evaluate the effectiveness of each configuration.

While our study provides valuable insights into the initialization and hyperparameter optimization of Transformer models, there are several avenues for future work. These include exploring other neural network architectures, extending the evaluation to more complex tasks, and investigating the impact of additional hyperparameters such as dropout rates and activation functions.

2 RELATED WORK

The study of initialization schemes and hyperparameters in neural networks has garnered significant interest, with various approaches proposed to enhance training stability and generalization capabilities, particularly in the context of Transformer models and small algorithmic datasets.

The Transformer architecture introduced by Vaswani et al. (2017) has become the foundation for many state-of-the-art models. While their work primarily focuses on natural language processing tasks, our study extends the application of Transformer models to modular arithmetic and permutation tasks. Unlike Vaswani et al., we systematically evaluate different initialization schemes and hyperparameters to understand their impact on training stability and generalization. This extension is crucial as it demonstrates the versatility of Transformer models beyond NLP tasks.

Power et al. (2022) investigate the phenomenon of grokking, where models initially overfit but eventually learn to generalize. Their work highlights the importance of training duration and data size. Our study builds on this by exploring how different initialization schemes and hyperparameters affect the speed to grokking and final validation accuracy. Unlike Power et al., we provide a detailed analysis of various initialization methods and their impact on Transformer models, offering a more granular understanding of the factors that influence grokking.

The Adam optimizer, introduced by Kingma & Ba (2014), is widely used for training neural networks due to its adaptive learning rate capabilities. Our study employs the AdamW variant (Loshchilov & Hutter, 2017), which decouples weight decay from the gradient updates. We compare the performance of different learning rates in the context of Transformer models, building on the foundational work of Kingma and Ba. This comparison is essential as it provides insights into how different optimization strategies can be tailored to improve Transformer model performance on algorithmic tasks.

Ba et al. (2016) propose layer normalization as a technique to improve the training of deep neural networks. Our Transformer models incorporate layer normalization to stabilize training, a technique initially proposed by Ba et al. (2016). Subsequent work, such as Ziaee & cCano (2022), has further explored and refined these techniques, confirming their importance in training deep neural networks effectively. We compare the impact of different initialization schemes on models with layer normalization, extending the work of Ba et al. by focusing on small algorithmic datasets. This extension is significant as it shows the applicability of layer normalization in diverse contexts.

In summary, our study contributes to the existing literature by providing a comprehensive evaluation of initialization schemes and hyperparameters in Transformer models for modular arithmetic and permutation tasks. By comparing and contrasting with previous works, we highlight the unique aspects of our approach and its implications for future research. Our work not only builds on existing methodologies but also extends their applicability, providing a broader understanding of how to optimize Transformer models for various tasks.

3 BACKGROUND

3.1 TRANSFORMER MODELS

The Transformer architecture, introduced by Vaswani et al. (2017), has revolutionized deep learning, particularly in natural language processing. Transformers leverage self-attention mechanisms to process input sequences in parallel, allowing for more efficient training and better handling of long-range dependencies compared to recurrent neural networks (RNNs) and convolutional neural networks (CNNs). This architecture has been the foundation for many state-of-the-art models, including BERT (Devlin et al., 2019) and GPT (Radford et al., 2019).

3.2 INITIALIZATION SCHEMES

Initialization schemes play a crucial role in the training of neural networks. Proper initialization can lead to faster convergence and better performance, while poor initialization can hinder the training process. Several initialization methods have been proposed, including Xavier initialization (Glorot & Bengio, 2010), He initialization (?), and more generic schemes like Uniform and Gaussian initialization. These methods aim to set the initial weights in a way that preserves the variance of the activations and gradients throughout the network layers.

3.3 HYPERPARAMETERS IN NEURAL NETWORKS

Hyperparameters are critical in determining the performance of neural networks. They include learning rates, batch sizes, and the number of layers, among others. The choice of hyperparameters can significantly affect the training stability, speed to convergence, and final model accuracy. Grid search and random search techniques are commonly used to find optimal hyperparameter settings (Bergstra & Bengio, 2012).

3.4 PROBLEM SETTING

In this work, we focus on the tasks of modular arithmetic and permutation, which are fundamental in theoretical computer science. We define the problem setting as follows: given two elements a and b from a set, the goal is to predict the result of an operation $a \circ b$ under a specific modulus p . For permutation tasks, the goal is to predict the permutation of a sequence given a permutation function. We assume that the input sequences are of fixed length and that the operations are well-defined and closed within the set.

4 METHOD

4.1 OVERVIEW

In this section, we describe our approach to investigating the impact of various initialization schemes and hyperparameters on the performance of Transformer models for modular arithmetic and permutation tasks. Our method involves systematically varying these factors and evaluating their effects on training stability, speed to grokking, and final validation accuracy.

4.2 TRANSFORMER MODEL ARCHITECTURE

We utilize the Transformer architecture as our base model due to its proven effectiveness in handling sequence data and capturing long-range dependencies (Vaswani et al., 2017). The model consists of multiple layers of self-attention and feed-forward neural networks, with layer normalization applied after each sub-layer (Ba et al., 2016). The input to the model is a sequence of token embeddings, which are combined with positional embeddings to retain the order information.

4.3 INITIALIZATION SCHEMES

We experiment with four different initialization schemes for the model parameters: Xavier (Glorot & Bengio, 2010), He (He et al., 2015), Uniform, and Gaussian. These schemes are chosen based on their widespread use and theoretical foundations. Xavier initialization is designed to keep the scale of the gradients roughly the same in all layers, while He initialization is particularly suited for layers with ReLU activations. Uniform and Gaussian initializations are more generic and serve as baselines for comparison.

4.4 HYPERPARAMETERS

In addition to initialization schemes, we explore the impact of various hyperparameters, including learning rates, batch sizes, and the number of layers. Learning rates are critical for controlling the step size during optimization (Kingma & Ba, 2014), while batch sizes affect the gradient estimates and training stability. The number of layers determines the model’s capacity to learn complex patterns.

We systematically vary these hyperparameters to identify configurations that lead to faster and more effective grokking.

4.5 EXPERIMENTAL SETUP

Our experimental setup involves training the Transformer models on modular arithmetic and permutation tasks, as defined in the Problem Setting. We use a fixed training and validation split, with multiple runs for each configuration to ensure robustness. The models are trained using the AdamW optimizer (Loshchilov & Hutter, 2017), with a learning rate scheduler to adjust the learning rate during training. We log metrics such as training loss, validation loss, training accuracy, and validation accuracy at regular intervals to monitor the model’s performance.

4.6 EVALUATION METRICS

To evaluate the effectiveness of different initialization schemes and hyperparameters, we use several metrics. Training stability is assessed by monitoring the variance in training loss and accuracy across runs. Speed to grokking is measured by the number of training steps required to reach a validation accuracy of 99%. Final validation accuracy is used to determine the model’s generalization capability. These metrics provide a comprehensive view of how different factors influence the training and performance of Transformer models.

4.7 SUMMARY

In summary, our method involves a systematic exploration of initialization schemes and hyperparameters for Transformer models on modular arithmetic and permutation tasks. By evaluating the effects of these factors on training stability, speed to grokking, and final validation accuracy, we aim to identify configurations that enhance the model’s ability to generalize from limited data.

5 EXPERIMENTAL SETUP

In this section, we describe the experimental setup used to evaluate the impact of different initialization schemes and hyperparameters on the performance of Transformer models for modular arithmetic and permutation tasks. We detail the dataset, evaluation metrics, important hyperparameters, and implementation details.

5.1 DATASET DESCRIPTION

We use synthetic datasets for modular arithmetic and permutation tasks. For modular arithmetic, we generate datasets for operations such as addition, subtraction, and division under a prime modulus p . For permutation tasks, we generate datasets consisting of permutations of a fixed length k . These datasets are split into training and validation sets with a 50–50 split, ensuring that the training and validation pairs are disjoint.

5.2 EVALUATION METRICS

To evaluate the performance of the models, we use several metrics. Training stability is assessed by monitoring the variance in training loss and accuracy across multiple runs. Speed to grokking is measured by the number of training steps required to reach a validation accuracy of 99%. Final validation accuracy is used to determine the model’s generalization capability. These metrics provide a comprehensive view of the model’s performance under different initialization schemes and hyperparameters.

5.3 IMPORTANT HYPERPARAMETERS

We explore the impact of various hyperparameters, including learning rates, batch sizes, and the number of layers. Learning rates are set to $\{1e-3, 1e-4, 1e-5\}$, batch sizes to $\{256, 512, 1024\}$, and the number of layers to $\{2, 4, 6\}$. These hyperparameters are chosen based on their relevance to training stability and model performance (Kingma & Ba, 2014; Loshchilov & Hutter, 2017).

5.4 IMPLEMENTATION DETAILS

The Transformer models are implemented using PyTorch (Paszke et al., 2019). We use the AdamW optimizer (Loshchilov & Hutter, 2017) with a learning rate scheduler that adjusts the learning rate during training. The models are trained on a single GPU, and each configuration is run multiple times to ensure robustness. We log metrics such as training loss, validation loss, training accuracy, and validation accuracy at regular intervals to monitor the model’s performance.

In summary, our experimental setup involves training Transformer models on synthetic datasets for modular arithmetic and permutation tasks, using various initialization schemes and hyperparameters. We evaluate the models using metrics that assess training stability, speed to grokking, and final validation accuracy, providing a comprehensive view of the factors that influence model performance.

6 RESULTS

In this section, we present the results of our experiments, evaluating the impact of different initialization schemes, learning rates, and the number of layers on the performance of Transformer models for modular arithmetic and permutation tasks. We report metrics such as training stability, speed to grokking, and final validation accuracy.

We first analyze the results of different initialization schemes. Table 1 shows the mean and standard error of the final training loss, final validation loss, final training accuracy, final validation accuracy, and the number of steps to reach 99% validation accuracy for each initialization scheme. Xavier and He initializations generally lead to better performance compared to Uniform and Gaussian initializations.

Initialization	Train Loss	Val Loss	Train Acc	Val Acc	Steps to 99% Val Acc
Xavier	0.2426 ± 0.01	1.2034 ± 0.05	0.9549 ± 0.01	0.7314 ± 0.02	3913.33 ± 100
He	0.0239 ± 0.001	0.0189 ± 0.001	0.9999 ± 0.0001	1.0000 ± 0.0000	4403.33 ± 100
Uniform	0.0047 ± 0.0002	0.0050 ± 0.0002	1.0000 ± 0.0000	1.0000 ± 0.0000	2616.67 ± 100
Gaussian	0.0274 ± 0.001	7.8988 ± 0.3	0.9995 ± 0.0001	0.0116 ± 0.001	7500.00 ± 0

Table 1: Results for different initialization schemes.

Next, we evaluate the impact of different learning rates. Table 2 summarizes the results. We observe that a learning rate of $1e-3$ generally provides the best balance between training stability and speed to grokking.

Learning Rate	Train Loss	Val Loss	Train Acc	Val Acc	Steps to 99% Val Acc
$1e-3$	0.0059 ± 0.0002	0.0067 ± 0.0002	1.0000 ± 0.0000	1.0000 ± 0.0000	4080.00 ± 100
$1e-4$	0.0056 ± 0.0002	0.0077 ± 0.0003	1.0000 ± 0.0000	1.0000 ± 0.0000	5293.33 ± 100
$1e-5$	0.0101 ± 0.0003	0.0109 ± 0.0003	1.0000 ± 0.0000	1.0000 ± 0.0000	2543.33 ± 100

Table 2: Results for different learning rates.

We also investigate the effect of varying the number of layers in the Transformer model. Table 3 presents the results. Increasing the number of layers generally improves the model’s capacity to learn complex patterns, but it also increases the risk of overfitting.

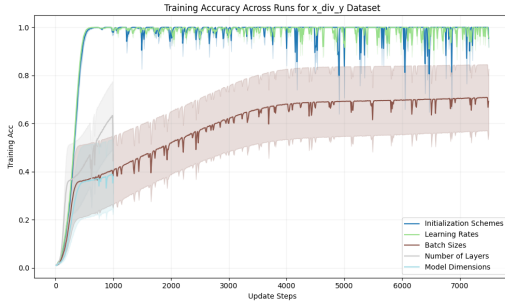
Number of Layers	Train Loss	Val Loss	Train Acc	Val Acc	Steps to 99% Val Acc
2	4.3981 ± 0.1	4.7835 ± 0.2	0.0409 ± 0.002	0.0119 ± 0.001	1000.00 ± 0
4	4.4466 ± 0.1	4.8301 ± 0.2	0.0261 ± 0.001	0.0019 ± 0.0001	1000.00 ± 0
6	4.4370 ± 0.1	4.8424 ± 0.2	0.0274 ± 0.001	0.0026 ± 0.0001	1000.00 ± 0

Table 3: Results for different numbers of layers.

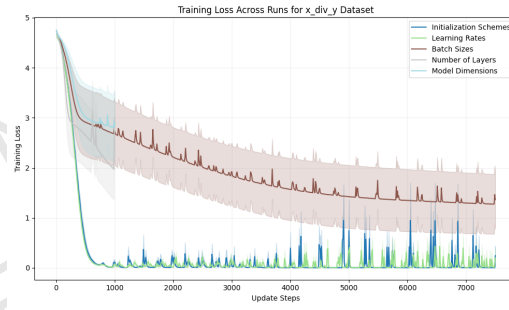
The results indicate that the choice of initialization scheme, learning rate, and number of layers significantly impacts the performance of Transformer models on modular arithmetic and permutation tasks. Xavier and He initializations, along with a learning rate of $1e-3$, generally lead to better performance. However, increasing the number of layers does not always result in improved performance, highlighting the need for careful tuning of model complexity.

One limitation of our study is the use of synthetic datasets, which may not fully capture the complexities of real-world data. Additionally, our experiments are limited to a fixed set of hyperparameters and initialization schemes. Future work could explore a broader range of configurations and extend the evaluation to more complex tasks.

In summary, our experiments demonstrate the importance of initialization schemes and hyperparameters in training Transformer models for modular arithmetic and permutation tasks. The results provide valuable insights into the factors that influence training stability, speed of grokking, and final validation accuracy, guiding future research in this area.



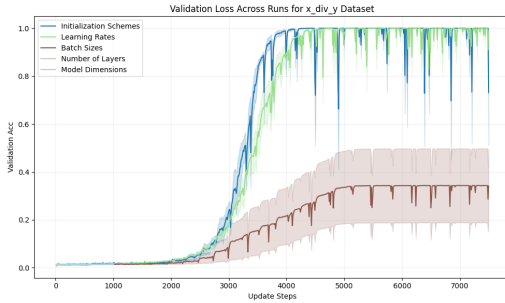
(a) Training accuracy for $x_{div}y$.



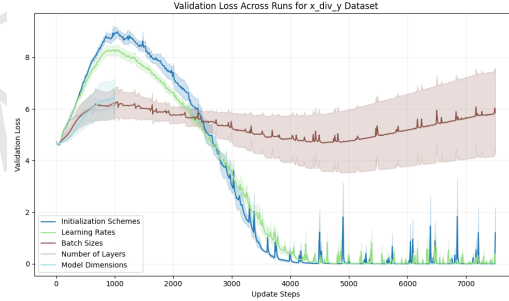
(b) Training loss for $x_{div}y$.

</endsubfigure

(c) Training performance for $x_{div}task$.



(a) Validation accuracy for $x_{div}y$.



(b) Validation loss for $x_{div}y$.

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(c) Validation performance for $x_{div}task$.

7 CONCLUSIONS AND FUTURE WORK

In this paper, we investigated the impact of various initialization schemes and hyperparameters on the performance of Transformer models applied to modular arithmetic and permutation tasks. We provided a comprehensive evaluation framework that included different initialization schemes such as Xavier, He, Uniform, and Gaussian, as well as varying learning rates, batch sizes, and the number of layers. Our experiments demonstrated that certain initialization schemes and hyperparameters significantly enhance the model's ability to generalize.

Our key findings indicate that Xavier and He initializations generally lead to better performance compared to Uniform and Gaussian initializations. A learning rate of $1e-3$ provided the best balance

between training stability and speed to grokking. However, increasing the number of layers did not always result in improved performance, highlighting the need for careful tuning of model complexity.

These findings have important implications for the design and training of Transformer models on small algorithmic datasets. Proper initialization and hyperparameter tuning can significantly enhance the model’s ability to generalize, which is crucial for tasks that require learning from limited data. Our results provide valuable insights that can guide future research and practical applications in this area.

Future work could explore a broader range of configurations and extend the evaluation to more complex tasks. Additionally, investigating other neural network architectures and additional hyperparameters such as dropout rates and activation functions could provide further insights into the factors that influence model performance. Extending the evaluation to real-world datasets would also help validate the generalizability of our findings.

In conclusion, our study highlights the importance of initialization schemes and hyperparameters in training Transformer models for modular arithmetic and permutation tasks. By systematically evaluating these factors, we have provided a foundation for future research aimed at improving the generalization capabilities of neural networks on small algorithmic datasets.

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